

VAPT INTERNSHIP

WEEK 4 Scanning and Enumeration

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1. SECTION A - Multiple Choice Questions

The image shows a Google Forms interface for a quiz titled "VAPT Internship - Nepal | Week 5 Quiz: Vulnerability assesment". The form is displayed in a web browser window. The header includes the user's email "sulavadh124@gmail.com" and a "Switch accounts" link. Below the header, there is a section titled "SECTION A - MCQs (20 Marks)". The quiz contains five multiple-choice questions related to Vulnerability Assessment (VA) and Penetration Testing. The first question asks for the primary goal of a VA, with the correct answer being "Identify and list vulnerabilities". The second question asks for the activity most associated with Penetration Testing, with the correct answer being "Active exploitation". The third question asks for the first phase in the VA methodology, with the correct answer being "Identify". The fourth question asks what the "Prioritize" phase involves, with the correct answer being "Assigning severity to vulnerabilities". The fifth question asks for the common scoring system for vulnerability severity, with the correct answer being "CVSS v3.1".

VAPT Internship - Nepal | Week 5
Quiz: Vulnerability assesment

sulavadh124@gmail.com [Switch accounts](#)
Not shared Draft saved

* Indicates required question

SECTION A - MCQs (20 Marks)

What is the primary goal of a Vulnerability Assessment (VA)? *

- ☐ A. Exploit vulnerabilities
- ☒ B. Identify and list vulnerabilities
- ☐ C. Destroy systems
- ☐ D. Develop malware

Which activity is most associated with Penetration Testing? *

- ☐ A. Passive scanning
- ☐ B. Risk classification
- ☒ C. Active exploitation
- ☐ D. Report formatting

In the VA methodology, which phase comes first? *

- ☐ A. Classify
- ☐ B. Report
- ☒ C. Identify
- ☐ D. Prioritize

What does the "Prioritize" phase mainly involve? *

- ☐ A. Writing reports
- ☒ B. Assigning severity to vulnerabilities
- ☐ C. Installing tools
- ☐ D. Removing vulnerabilities

Which scoring system is commonly used to measure vulnerability severity? *

- ☐ A. ISO
- ☒ B. CVSS v3.1
- ☐ C. TCP/IP
- ☐ D. OWASP Top 10

New tab VAPI Internship — Nepal | Weebly

docs.google.com/forms/d/e/1FAIpQLSebkF4jKzOWIs8t863gp2qPu-8b9BJonlkbFp_rdPi8CUgUg/formResponse?pli=1

☐ A. Time-based changes

☐ B. Environmental factors

☒ C. Intrinsic characteristics of a vulnerability

☐ D. User behavior

Which tool is associated with vulnerability scanning? *

☐ A. Metasploit

☐ B. Wireshark

☒ C. OpenVAS

☐ D. Burp Suite

What is the role of Greenbone in OpenVAS? *

☐ A. Malware detection

☐ B. Network routing

☒ C. Management and reporting

☐ D. Encryption

What does CVE stand for? *

☐ A. Common Vulnerability Enumeration

☒ B. Common Vulnerabilities and Exposures

☐ C. Critical Vulnerability Evaluation

☐ D. Cyber Vulnerability Engine

What does CVE stand for? *

☐ A. Common Vulnerability Enumeration

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☐ D. Cyber Vulnerability Engine

The NVD is maintained by which organization? *

☐ A. Microsoft

☐ B. Google

☒ C. NIST

☐ D. ISO

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Google Forms

2. SECTION B - Short Answer Questions

Explain the CVSS v3.1 scoring system in detail. (5Marks) *

CVSS (Common Vulnerability Scoring System) v3.1 is a standard method used to measure the severity of security vulnerabilities. It helps organizations understand how serious a vulnerability is and prioritize remediation.

CVSS scores range from 0.0 to 10.0:
 0.1–3.9: Low
 4.0–6.9: Medium
 7.0–8.9: High
 9.0–10.0: Critical

CVSS has three metric groups:

1. Base Metrics
 These measure the intrinsic characteristics of a vulnerability that do not change over time.
 Examples:
 -Attack Vector
 -Attack Complexity
 -Privileges Required
 -User Interaction
 -Confidentiality Impact
 -Integrity Impact
 -Availability Impact
2. Temporal Metrics
 These measure factors that can change over time, such as:
 -Availability of exploit code
 -Patch availability
 -Confidence in the vulnerability report
3. Environmental Metrics
 These consider the specific environment of an organization, such as:
 -Importance of the affected system
 -Security requirements of the business

Discuss the role of OpenVAS/Greenbone in vulnerability assessment. (5Marks) *

OpenVAS (Open Vulnerability Assessment System) is an open-source vulnerability scanning tool used to detect security weaknesses in networks and systems. It helps organizations perform regular vulnerability assessments.

Greenbone is the company and framework that manages and supports OpenVAS. It provides the management interface, reporting features, and vulnerability feed updates.

Role in Vulnerability Assessment:

- Scans systems for known vulnerabilities
- Detects missing patches and outdated software
- Identifies misconfigurations and insecure services
- Uses regularly updated vulnerability databases (feeds)
- Generates detailed reports with severity ratings
- Helps prioritize remediation using CVSS scoring

Importance:
 OpenVAS/Greenbone automates vulnerability scanning, saves time, improves accuracy, and helps organizations maintain stronger security by continuously identifying and managing risks.

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3. SECTION C - Practical Submission Tasks

a. Task C1 - CVE Research Table

Service & Version	CVE ID	CVSS Score	Severity	What the Attacker Can Do
vsftpd 2.3.4 (port 21)	CVE-2011-2523	10.0	Critical	Attacker can use the backdoor by sending :) in the username and gain unauthenticated root shell access remotely.
OpenSSH 4.7p1 (port 22)	CVE-2008-4109	5.0	Medium	Attacker may exploit weak cryptographic handling and perform information disclosure or

				session attacks under specific conditions.
Apache 2.2.8 (port 80)	CVE-2011-3192	7.8	High	Attacker can send crafted HTTP Range headers causing Denial of Service (DoS) and crash the web server.
MySQL 5.0.51a (port 3306)	CVE-2012-2122	10.0	Critical	Attacker can bypass MySQL authentication and log in without valid credentials, leading to full database compromise.
Samba smbd 3.0.20 (port 445)	CVE-2007-2447	10.0	Critical	Attacker can execute arbitrary commands remotely using the username map script vulnerability and gain system-level access.

b. Task C2- CVSS Scoring Exercise

CVSS Metric	Your Selection	Reason
Attack Vector (AV)	Network (N)	Exploitable remotely through the internet
Attack Complexity (AC)	Low (L)	No special conditions needed
Privileges Required (PR)	None (N)	No login required
User Interaction (UI)	None (N)	No victim action needed
Scope (S)	Unchanged (U)	Only the database is affected
Confidentiality Impact (C)	High (H)	Entire user database is exposed
Integrity Impact (I)	None (N)	No data modification mentioned
Availability Impact (A)	None (N)	No service disruption mentioned

The screenshot shows the first.org CVSS calculator interface. The Base Score is 7.5 (High). The selected metrics are:

- Attack Vector (AV): Network (N)
- Attack Complexity (AC): Low (L)
- Privileges Required (PR): None (N)
- User Interaction (UI): None (N)
- Scope (S): Unchanged (U)
- Confidentiality (C): High (H)
- Integrity (I): None (N)
- Availability (A): None (N)

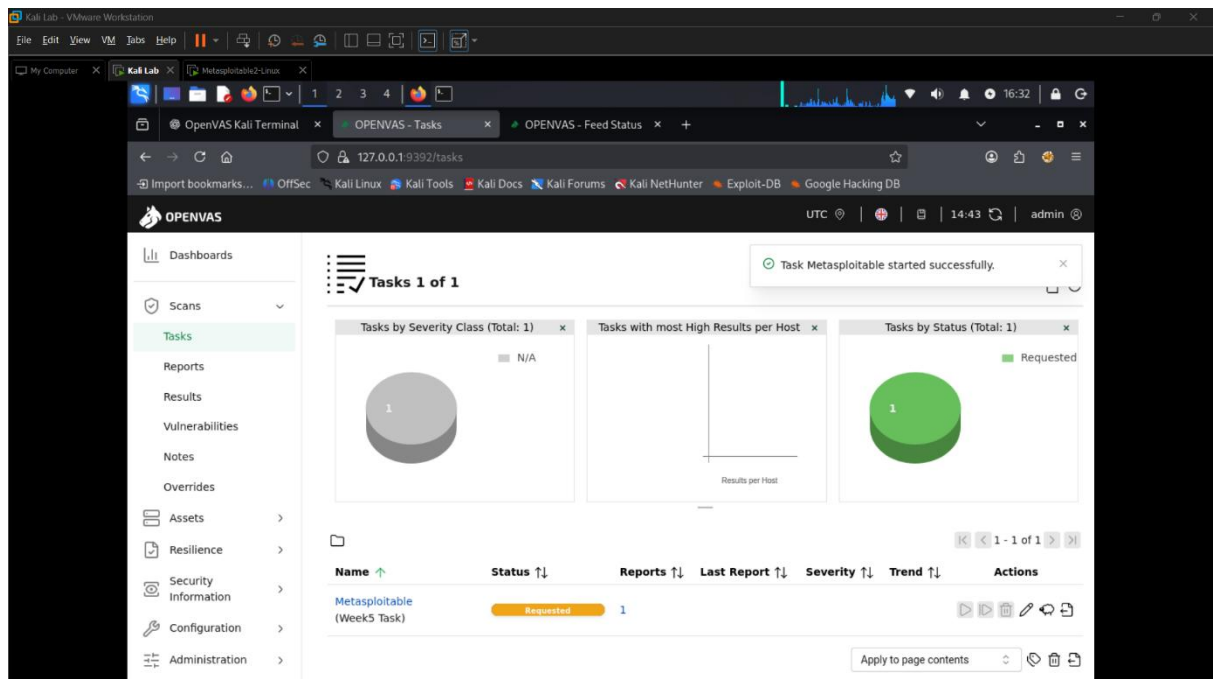
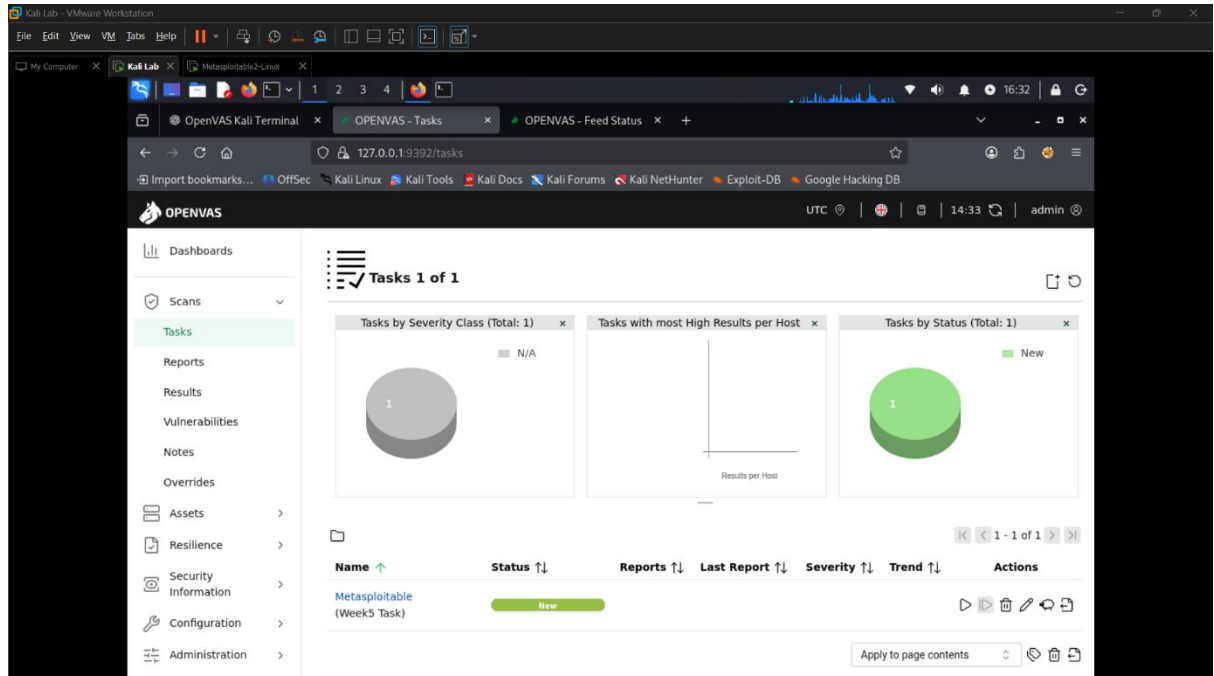
The Vector String is: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:N/A:N

The Temporal Score is also displayed at the bottom right.

c. OpenVAS Scan Screenshots

Screenshot 1 - Scan Status:

The vulnerability scan was performed on the target system (192.168.63.128) using OpenVAS. The scan task was successfully created and executed. The status of the scan shows that it is either running or completed, indicating that the scanning process is functioning correctly.



Kali Lab - VMware Workstation

File Edit View VM Tabs Help

My Computer Kali Lab Metasploit2-Linux

OpenVAS Kali Terminal OPENVAS - Tasks OPENVAS - Feed Status

127.0.0.1:9392/tasks

Import bookmarks... OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

OPENVAS UTC 14:30 admin

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Results

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Assets

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Security Information

Configuration

Administration

Tasks 1 of 1

Tasks by Severity Class (Total: 1)

Tasks with most High Results per Host

Tasks by Status (Total: 1)

Queued

1 - 1 of 1

Name	Status	Reports	Last Report	Severity	Trend	Actions
Metasploitable (Week 5 Task)	Queued	1				

Apply to page contents

Kali Lab - VMware Workstation

File Edit View VM Tabs Help

My Computer Kali Lab Metasploit2-Linux

GVM setup and UI issues New Tab OPENVAS - Tasks OPENVAS - Feed Status

127.0.0.1:9392/tasks

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OPENVAS UTC 4:56 admin

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Tasks 1 of 1

Tasks by Severity Class (Total: 1)

Tasks with most High Results per Host

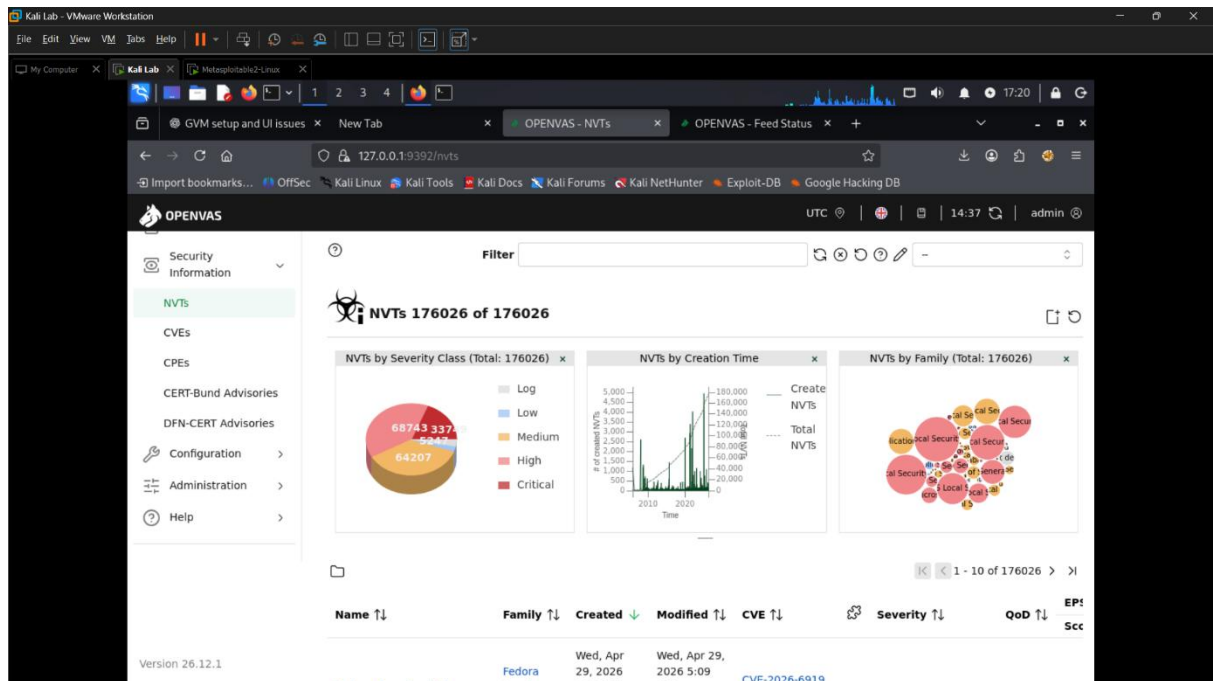
Tasks by Status (Total: 1)

Done

1 - 1 of 1

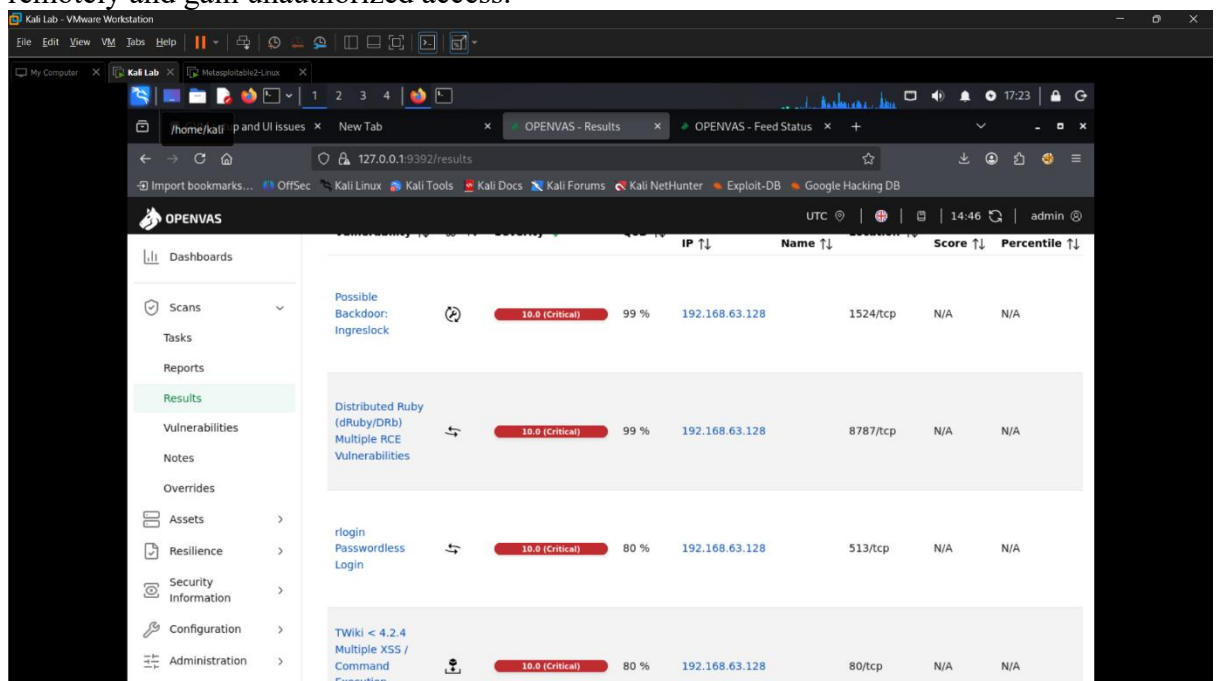
Name	Status	Reports	Last Report	Severity	Trend	Actions
Metasploitable (Week 5 Task)	Done	1	Thu, Apr 30, 2026 11:01 AM Coordinated Universal Time	10.0 (Critical)		

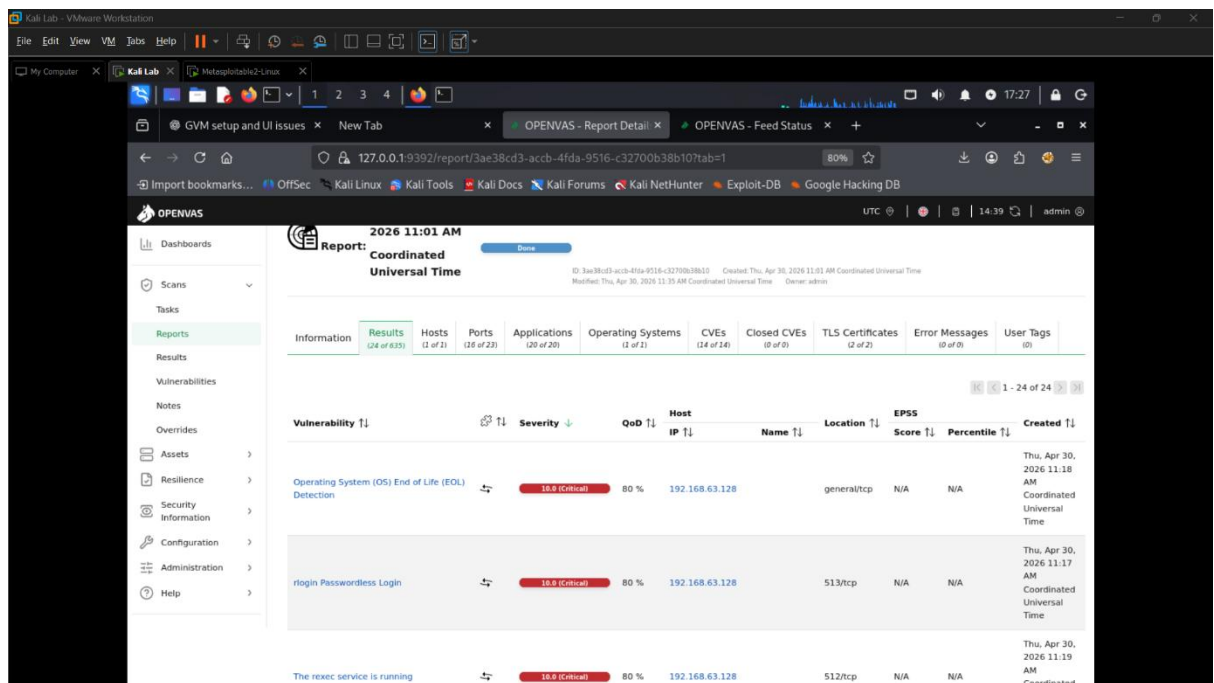
Apply to page contents



Screenshot 2 - High/Critical Findings

The scan results display multiple vulnerabilities detected on the target system. Several vulnerabilities are classified as **High** and **Critical**, indicating serious security risks. These vulnerabilities may allow attackers to exploit the system remotely and gain unauthorized access.

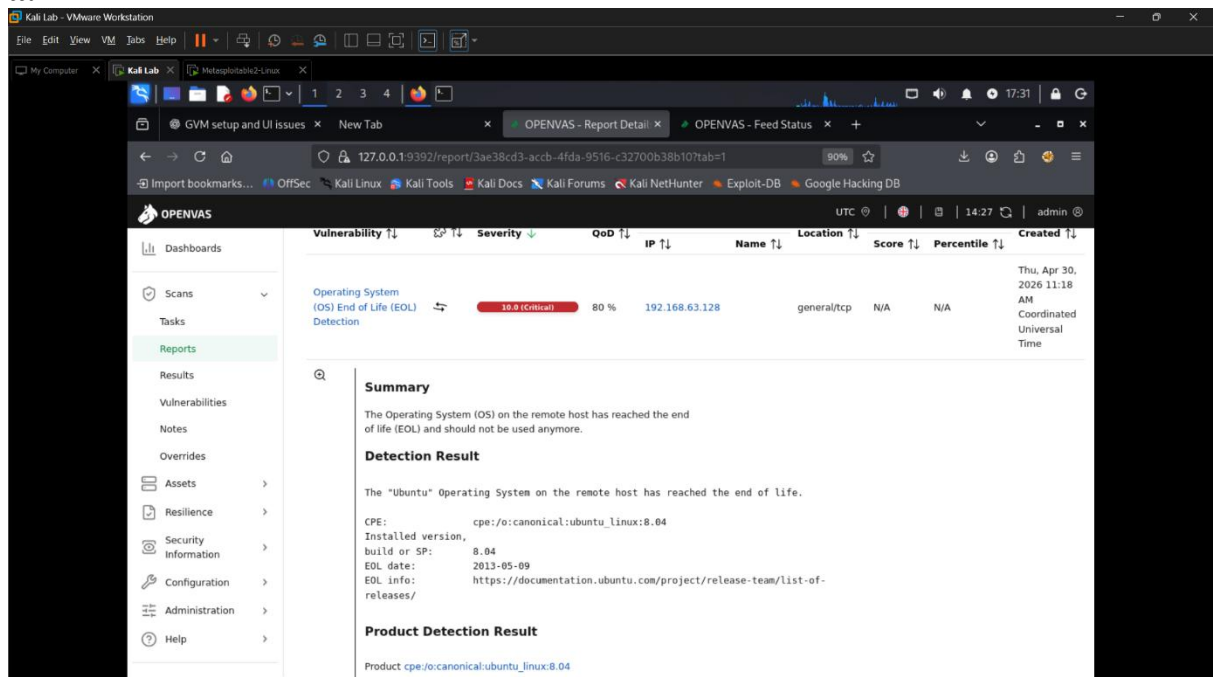


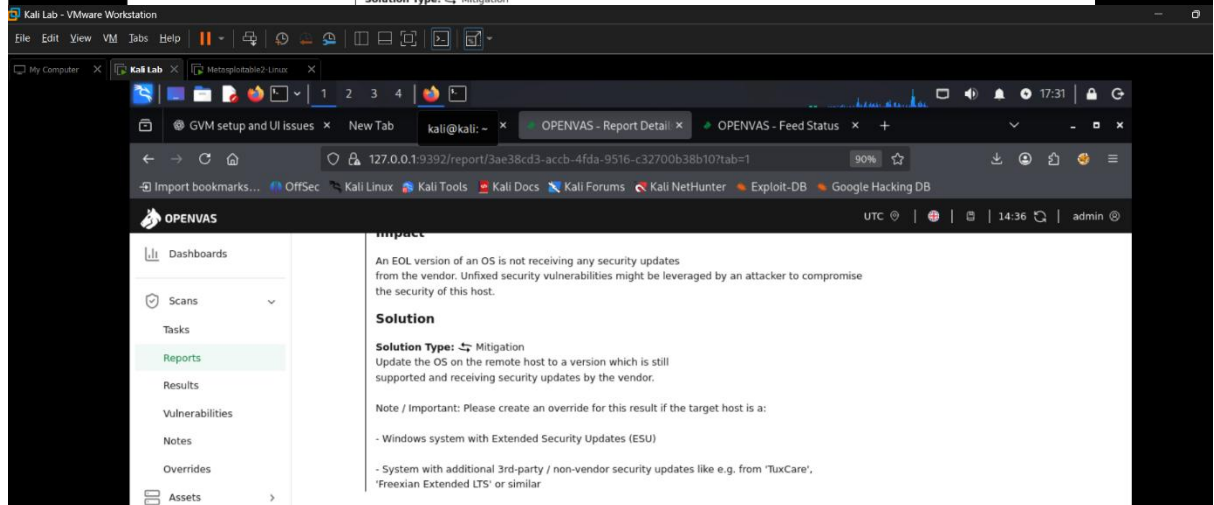
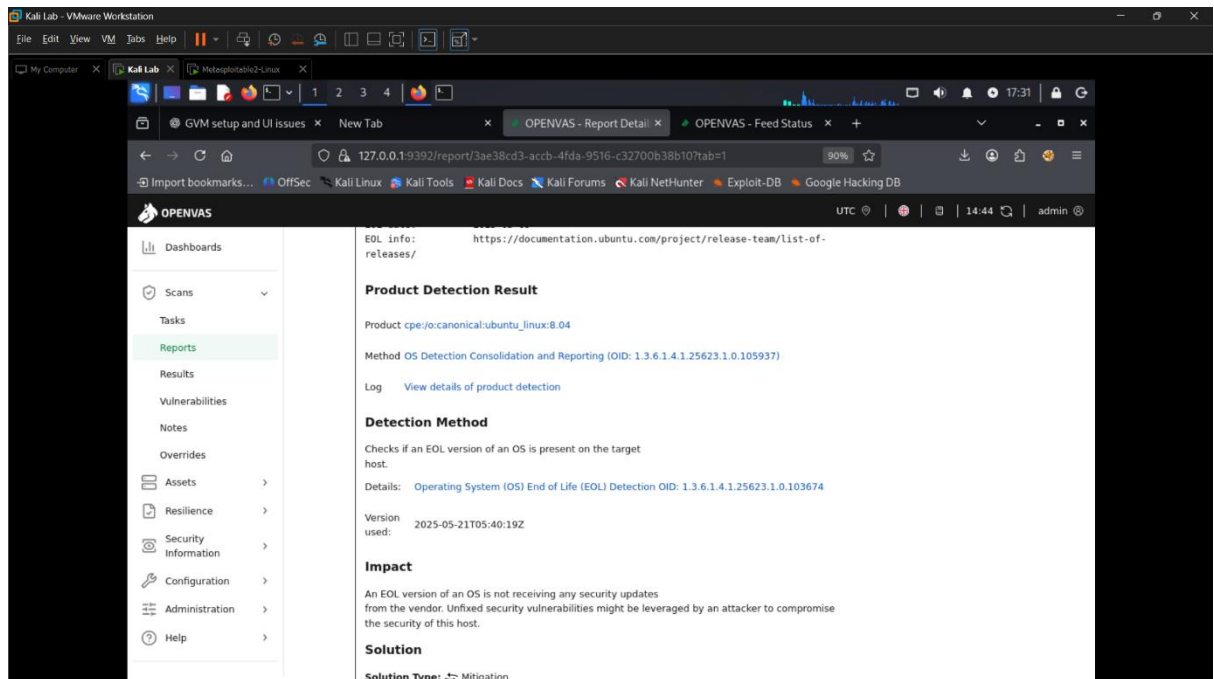


Screenshot 3 - Vulnerability Detail

The detailed vulnerability report provides information such as the **CVE ID**, **CVSS score**, and description of the vulnerability. The selected vulnerability has a high CVSS score, indicating severe impact. It also shows the affected host (192.168.63.128) and possible security risks associated with it

a.





b.

The screenshot shows the OpenVAS web interface in a browser window. The left sidebar contains a navigation menu with options like Dashboards, Scans, Tasks, Reports, Results, Vulnerabilities, Notes, Overrides, Assets, Resilience, Security Information, Configuration, Administration, and Help. The main content area displays the 'Report Detail' for a vulnerability titled 'rlogin Passwordless Login'. The report includes a summary, detection result, detection method, impact, and solution. The detection result states: 'It was possible to gain root access without a password.' The detection method describes a check for a vulnerable version of the rlogin service. The impact states: 'This vulnerability allows an attacker to gain complete control over the target system.' The solution suggests disabling the rlogin service and using alternatives like SSH instead.

Summary

The rlogin service allows root access without a password.

Detection Result

It was possible to gain root access without a password.

Detection Method

Checks if a vulnerable version is present on the target host.

Details: [rlogin Passwordless Login OID: 1.3.6.1.4.1.25623.1.0.113766](#)

Version used: 2020-09-30T09:30:12Z

Impact

This vulnerability allows an attacker to gain complete control over the target system.

Solution

Disable the rlogin service and use alternatives like SSH instead.

c.

The screenshot shows the OpenVAS web interface in a browser window. The left sidebar contains a navigation menu with options like Dashboards, Scans, Tasks, Reports, Results, Vulnerabilities, Notes, Overrides, Assets, Resilience, Security Information, Configuration, Administration, and Help. The main content area displays the 'Report Detail' for a vulnerability titled 'TWiki < 4.2.4 Multiple XSS / Command Execution Vulnerabilities'. The report includes a summary, detection result, insight, and detection method. The detection result states: 'TWiki is prone to multiple cross-site scripting (XSS) and command execution vulnerabilities.' The insight describes the flaws: '%URLPARAM{}% variable is not properly sanitized which lets attackers conduct cross-site scripting attack' and '%SEARCH{}% variable is not properly sanitised before being used in an eval() call which lets the attackers execute perl code through eval injection attack.' The detection method describes the check for a vulnerable version of TWiki.

Summary

TWiki is prone to multiple cross-site scripting (XSS) and command execution vulnerabilities.

Detection Result

Installed version: 01.Feb.2003
Fixed version: 4.2.4

Insight

The flaws are due to:

- %URLPARAM{}% variable is not properly sanitized which lets attackers conduct cross-site scripting attack.
- %SEARCH{}% variable is not properly sanitised before being used in an eval() call which lets the attackers execute perl code through eval injection attack.

Detection Method

Checks if a vulnerable version is present on the target host.

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File Edit View VM Tabs Help

My Computer Kali Lab Mutasploit62-Linux

1 2 3 4

/home/kali/p and UI issues New Tab

OPENVAS - Report Detail x OPENVAS - Feed Status x

127.0.0.1:9392/report/3ae38cd3-accb-4fda-9516-c32700b38b10?tab=1 90%

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OPENVAS UTC 14:40 admin

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Help

Impact

Successful exploitation could allow execution of arbitrary script code or commands. This could let attackers steal cookie-based authentication credentials or compromise the affected application.

Solution

Solution Type: Vendorfix

Update to version 4.2.4 or later.

References

CVE CVE-2008-5304 CVE-2008-5305

Other <http://twiki.org/cgi-bin/view/Codev/SecurityAlert-CVE-2008-5304> <http://www.securityfocus.com/bid/32668> <http://www.securityfocus.com/bid/32669> <http://twiki.org/cgi-bin/view/Codev/SecurityAlert-CVE-2008-5305>

Distributed Ruby (dRuby/DRb) Multiple RCE Vulnerabilities	10.0 (Critical)	99 %	192.168.63.128	8787/tcp	N/A	N/A	Thu, Apr 30, 2026 11:22 AM Coordinated Universal Time
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